A-A-51183
19 June 1990

COMMERCIAL ITEM DESCRIPTION

POWER SOURCE, ARC, DC, CONSTANT CURRENT, FOR SHIELDED METAL ARC WELDING (SMAW)

The General Services Administration has authorized the use of this commercial item description in preference to MIL-W-80043.

Abstract. This description covers direct current (DC) arc welding power sources delivering a rated constant output current at 60 percent duty cycle. The power sources are used most commonly for single operator shielded metal arc, or "stick," welding (SMAW).

Classification. The power source shall be of the following sizes. The size and quantity to be furnished shall be as specified (see ordering data).

Size	Current (amperes)
200	200
300	300
400	400
500	500
600	600

NOTE: "Size" is defined as the output expressed as rated load current when the power source is operated at rated input voltage, rated duty cycle, and rated frequency.

Beneficial comments, recommendations, additions, deletions, clarifications,or any data which may improve this document should be sent to: Defense Industrial Plant Equipment Center, ATTN: DIPEC-SSM, 2163 Airways Boulevard, Memphis, TN 38114-5051.

AMSC N/A FSC 3431

DISTRIBUTION STATEMENT A. Approved for public release; distribution is unlimited.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T08:52Z
Check the source to verify that this is the current version before use.

Salient characteristics. The welding power source shall meet the following requirements. All requirements specified in this document are the minimum requirements of the Government and are not intended to limit the manufacturer in machine design or performance.

1. The power source shall conform to the requirements of NEMA EW 1 for NEMA Class I (60) constant current power sources.

2. The power source shall be wired to conform to the requirements of NEMA EW 1 and NFPA 70. Unless otherwise specified (see ordering data), the power source shall be designed to operate on 3-phase, 60 Hertz (Hz), dual 230/460 volts. Conversion from one voltage to another shall be accomplished by the reconnection of existing components. The power source shall be initially wired for operation the voltage specified (see ordering data).

3. Each power source shall be provided with protection devices to prevent damage to power source components due to excessive temperature, transient voltages, and electrical overload.

4. Each power source shall be provided with line voltage and load compensation. A line voltage fluctuation of ± 10 percent shall not adversely affect the output of the power source.

5. Each power source shall be provided with a power switch; a pilot light to indicate the power source is on; and a weld current control covering the full current range for the machine.

6. The power source shall be portable and mounted on a base. Each power source shall be provided with means for lifting at a designated point by crane or hoisting mechanism equipped with a hook, shackle, or clevis and for handling by fork lift truck. Running gear, if required, shall be as specified (see ordering data).

7. The power source shall operate in parallel with two or more similar power sources of the same manufacture when connected according to the manufacturer's instructions.

8. The following optional equipment shall be furnished when specified and fully described (see ordering data). The equipment shall be functional without requiring modification of the power source or the equipment.

a. Voltmeter and ammeter (or voltmeter/ammeter combination) to measure the welding power output of the power source.

b. Polarity reversing switch to change from one polarity to another without removing the leads from the terminals.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T08:52Z
Check the source to verify that this is the current version before use.

c. Remote hand current control to permit current adjustment without returning to the power source with a minimum of 20 feet of cable.

Contractor certification. The contractor shall certify and maintain substantiating evidence that the product offered meets the salient characteristics of this commercial item description, and that the product conforms to the producer's own drawings, specifications, standards, and quality assurance practices. The Government reserves the right to require proof of such conformance prior to first delivery and thereafter as may be otherwise provided for under the provisions of the contract.

Metric products. Products manufactured to metric dimensions will be considered on an equal basis with those manufactured using inch-pound units, provided they fall within specified tolerances using conversions tables contained in the latest revision of FED-STD-376, and all other requirements of this commercial item description are met.

If a product is manufactured to metric dimensions and those dimensions exceed the tolerances specified in the inch-pound units, a request shall be made to the contracting officer to determine if the product is acceptable.

The contracting officer has the option of accepting or rejecting the product.

Regulatory requirements. In accordance with section 23.403 of the Federal Acquisition Regulations, the Government's policy is to acquire items composed of the highest percentage of recovered materials practicable, consistent with maintaining a satisfactory level of competition without adversely affecting performance requirements or exposing suppliers' employees to undue hazards from the recovered materials.

Environmental protection. The power source shall meet all applicable Environmental Protection Agency (EPA) and OSHA restrictions for materials classified as hazardous to the environment in effect on the date of the contract. Specific environmental protection requirements exceeding the established EPA and OSHA regulations as required herein shall be as specified by the procuring activity in the ordering data.

Preservation, packaging, packing, and marking. Preservation, packaging, packing, labeling, and marking shall be as specified in the contract or purchase order.

CID based part identification number. The following part identification numbering procedure is for Government purposes and does not constitute a requirement for the contractor.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T08:52Z
Check the source to verify that this is the current version before use.

A51183 XXX

Size

CID Number

Notes.

Ordering data. Purchasers should select the preferred options permitted herein and include the following information in procurement documents:

a. Title, number, and date of this commercial item description.

b. Size and quantity required.

c. Electrical requirements, if different.

d. Input voltage required.

e. Running gear, if required.

f. Optional equipment required; specify and fully describe.

g. Other requirements.

Federal Standard FED-STD-376, Preferred Metric Units for Government Use, is available from the Standardization Documents Order Desk, Building 4D, 700 Robbins Avenue, Philadelphia, PA 19111-5099.

NEMA EW 1, Electric Arc Welding Power Sources, is available from the National Electrical Manufacturers' Association, 2101 L Street, N.W., Washington, DC 20037-1580.

NFPA 70 , National Electrical Code, is available from the National Fire Protection Association, Batterymarch Park, Quincy, MA 02269-9101.

Sources of supply. The following manufacturers offer products which were known to meet the requirements of this CID at the time it was prepared. However, this in no way limits competition to those manufacturers listed.

Hobart Brothers Company
Lincoln Electric Company
L-Tec Welding and Cutting Systems
Miller Electric Manufacturing Company
PowCon

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T08:52Z
Check the source to verify that this is the current version before use.

Electrode holders for shielded metal arc welding are covered by commercial item description A-A-51161, Holder Electrode, Shielded Metal Arc Welding.

MILITARY INTERESTS:

Custodians

Army - AL
Navy - SH
Air Force - 99

Review Activities

Air Force - 84
Navy - YD, CG
Army - ME
DLA - GS

User Activities

Navy - MC, AS, ME

CIVIL AGENCY COORDINATION ACTIVITY:

GSA - FSS

PREPARING ACTIVITY:

DLA - IP

Project 3431-0208

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T08:52Z
Check the source to verify that this is the current version before use.